Ordering Fractions and Mixed Numbers

Answer Key

Grades 4–5

Quick Answers

1. $\frac{3}{4}$
2. $\frac{7}{3}$
3. —
4. $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$
5. —
6. $\frac{11}{4}$
7. $\frac{5}{2}$, $\frac{19}{8}$, $\frac{9}{4}$
8. —
9. $\frac{5}{6}$
10. $\frac{1}{2}$, $\frac{3}{5}$, $\frac{13}{20}$, $\frac{7}{10}$
11. —
12. $\frac{11}{8}$, $\frac{3}{2}$, $\frac{5}{3}$, $\frac{11}{6}$

1. Rewrite $\frac{3}{4}$ with denominator 12.

Step 1. Ask what the old denominator was multiplied by to reach the new one: $4 \times 3 = 12$, so the scale factor is 3.
Step 2. An equivalent fraction keeps the same value only if the numerator is scaled by the *same* factor: $3 \times 3 = 9$.
Step 3. $\frac{3}{4} = \boxed{\dfrac{9}{12}}$

2. Write $2\frac{1}{3}$ as a fraction greater than one.

Step 1. The whole number counts complete groups of thirds: each whole is $\frac{3}{3}$, so 2 wholes are $\frac{6}{3}$.
Step 2. Add the fractional part: $\frac{6}{3} + \frac{1}{3}$, which keeps the denominator and adds the numerators.
Step 3. $2\frac{1}{3} = \boxed{\dfrac{7}{3}}$

3. Compare $\frac{3}{8}$ and $\frac{1}{2}$.

Step 1. Use the half benchmark — half of 8 is 4, so $\frac{4}{8}$ is exactly one half.
Step 2. The numerator 3 is less than 4, so $\frac{3}{8}$ sits below the half mark while $\frac{1}{2}$ is on it.
Step 3. $\boxed{\frac{3}{8} < \frac{1}{2}}$

4. Order $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{8}$ least to greatest.

Step 1. All three denominators divide 8, so eighths is the common form — no benchmark needed once they match.
Step 2. $\frac{1}{2} = \frac{4}{8}$ and $\frac{1}{4} = \frac{2}{8}$; $\frac{3}{8}$ is already there.
Step 3. Order the numerators $2 < 3 < 4$, then write the numbers back in their original form: $\boxed{\frac{1}{4} < \frac{3}{8} < \frac{1}{2}}$

5. Compare $\frac{7}{8}$ and $\frac{5}{6}$.

Step 1. Both are just below one, so the half benchmark is useless — compare the *gap to one* instead.
Step 2. $\frac{7}{8}$ is $\frac{1}{8}$ short of one; $\frac{5}{6}$ is $\frac{1}{6}$ short of one.
Step 3. With equal numerators, the larger denominator is the smaller piece: $\frac{1}{8} < \frac{1}{6}$. The number with the smaller gap is closer to one, hence larger: $\boxed{\frac{7}{8} > \frac{5}{6}}$
Check with a common denominator (24): $\frac{21}{24}$ against $\frac{20}{24}$ — same conclusion.

6. Write $\frac{11}{4}$ as a mixed number.

Step 1. A fraction greater than one hides whole groups: ask how many complete fourths-groups of 4 fit inside 11.
Step 2. $11 \div 4 = 2$ with remainder 3, so there are 2 wholes and 3 fourths left over.
Step 3. $\frac{11}{4} = \boxed{2\,\frac{3}{4}}$

7. Order $2\frac{1}{2}$, $\frac{9}{4}$, $2\frac{3}{8}$ greatest to least.

Step 1. Two are mixed numbers and one is not, so convert the odd one out: $\frac{9}{4} = 2\frac{1}{4}$. Now every number has whole part 2, so only the fraction parts decide the order.
Step 2. Put the fraction parts over eighths: $\frac{1}{2} = \frac{4}{8}$, $\frac{3}{8}$ stays, $\frac{1}{4} = \frac{2}{8}$.
Step 3. Greatest to least by numerator, $4 > 3 > 2$, written back in the given forms: $\boxed{2\,\frac{1}{2} > 2\,\frac{3}{8} > \frac{9}{4}}$

8. Compare $\frac{5}{9}$ and $\frac{4}{7}$.

Step 1. Both are just above one half ($\frac{4.5}{9}$ and $\frac{3.5}{7}$ are the halves), so the benchmark ties and a common denominator is the reliable route.
Step 2. $9 \times 7 = 63$ works as a common denominator: $\frac{5}{9} = \frac{35}{63}$ and $\frac{4}{7} = \frac{36}{63}$.
Step 3. $35 < 36$, so $\boxed{\frac{5}{9} < \frac{4}{7}}$

9. Write $\frac{5}{6}$ with denominator 30.

Step 1. $6 \times 5 = 30$, so the scale factor is 5.
Step 2. Scale the numerator by the same factor: $5 \times 5 = 25$.
Step 3. $\frac{5}{6} = \boxed{\dfrac{25}{30}}$

10. Order $\frac{3}{5}$, $\frac{1}{2}$, $\frac{7}{10}$, $\frac{13}{20}$ least to greatest.

Step 1. All four denominators divide 20, so twentieths is the one common form that needs no guessing.
Step 2. $\frac{3}{5} = \frac{12}{20}$, $\frac{1}{2} = \frac{10}{20}$, $\frac{7}{10} = \frac{14}{20}$, and $\frac{13}{20}$ is already there.
Step 3. The numerators order as $10 < 12 < 13 < 14$, so $\boxed{\frac{1}{2} < \frac{3}{5} < \frac{13}{20} < \frac{7}{10}}$

11. Compare $1\frac{3}{4}$ and $\frac{5}{3}$.

Step 1. The forms differ, so put both in one form. $\frac{5}{3} = 1\frac{2}{3}$, and now both have whole part 1.
Step 2. Compare the fraction parts $\frac{3}{4}$ and $\frac{2}{3}$ over twelfths: $\frac{9}{12}$ against $\frac{8}{12}$.
Step 3. $9 > 8$, so $\boxed{1\,\frac{3}{4} > \frac{5}{3}}$

12. Challenge: order $\frac{5}{3}$, $1\frac{1}{2}$, $\frac{11}{8}$, $1\frac{5}{6}$ least to greatest.

Step 1. Three forms are in play, so pick ONE common form. Every number is between one and two, so mixed numbers with a common fraction denominator is the shortest route; the denominators 3, 2, 8, 6 all divide 24.
Step 2. Convert to mixed numbers first: $\frac{5}{3} = 1\frac{2}{3}$ and $\frac{11}{8} = 1\frac{3}{8}$; the other two are already mixed. Every whole part is 1, so only the fraction parts decide.
Step 3. Over twenty-fourths: $\frac{2}{3} = \frac{16}{24}$, $\frac{1}{2} = \frac{12}{24}$, $\frac{3}{8} = \frac{9}{24}$, and $\frac{5}{6} = \frac{20}{24}$.
Step 4. Ordering those numerators gives $9 < 12 < 16 < 20$, so in the original forms: $\boxed{\frac{11}{8} < 1\,\frac{1}{2} < \frac{5}{3} < 1\,\frac{5}{6}}$